```
from google.colab import drive
drive.mount('/content/drive/')
```

```
Mounted at /content/drive/
```

```
!pip install spectral
```

```
Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
Collecting spectral
  Downloading spectral-0.23.1-py3-none-any.whl (212 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 212.9/212.9 KB 5.6 MB/s eta 0:00:00
Requirement already satisfied: numpy in /usr/local/lib/python3.9/dist-packages (from spectral) (1.22.4)
Installing collected packages: spectral
Successfully installed spectral-0.23.1
```

```
%tensorflow_version 2.x
import tensorflow as tf
import random
from tensorflow.keras import Sequential,layers
from tensorflow.keras.models import Model
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import ModelCheckpoint
import statistics
from sklearn.decomposition import PCA
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report, cohen_kappa_score
import tensorflow_probability as tfp
from operator import truediv
from tensorflow.compat.v1.distributions import Bernoulli
from plotly.offline import init_notebook_mode

import numpy as np
import matplotlib.pyplot as plt
import scipy.io as sio
import os
import spectral
import tensorflow as tf
from tensorflow.python.keras import backend as K

init_notebook_mode(connected=True)
%matplotlib inline
```

```
Colab only includes TensorFlow 2.x; %tensorflow_version has no effect.
```

```
# Global variables
im_width, im_height, im_depth, im_channel = 11,11,30, 1
mc_loss_weight = 5
```

Model Construction

```
def calc_euclidian_dists(x, y):
  # x : (n,d)
  # y : (m,d)
    n = x.shape[0]
    m = y.shape[0]
    x = tf.tile(tf.expand_dims(x, 1), [1, m, 1])
    y = tf.tile(tf.expand_dims(y, 0), [n, 1, 1])
    return tf.reduce_mean(tf.math.pow(x - y, 2), 2)
```

```
def _bernoulli(shape, mean):
    return tf.nn.relu(tf.sign(mean - tf.random.uniform(shape, minval=0, maxval=1, dtype=tf.float32)))
```

```
class DropBlock3D(tf.keras.layers.Layer):
    def __init__(self, keep_prob, block_size, scale=True, **kwargs):
        super(DropBlock3D, self).__init__(**kwargs)
        self.keep_prob = float(keep_prob) if isinstance(keep_prob, int) else keep_prob
        self.block_size = int(block_size)
        self.scale = tf.constant(scale, dtype=tf.bool) if isinstance(scale, bool) else scale

    def compute_output_shape(self, input_shape):
        return input_shape

    def build(self, input_shape):
        assert len(input_shape) == 5
        _, self.d, self.h, self.w, self.channel = input_shape.as_list()
```

```
        # pad the mask
        p0 = (self.block_size - 1) // 2
        p1 = (self.block_size - 1) - p0
        self.padding = [[0, 0], [p0, p1], [p0, p1], [p0, p1], [0, 0]]
        self.set_keep_prob()
        super(DropBlock3D, self).build(input_shape)

    def call(self, inputs, training=True, **kwargs):
        def drop():
            mask = self._create_mask(tf.shape(inputs))
            output = inputs * mask
            output = tf.cond(self.scale,
                             true_fn=lambda: output * tf.compat.v1.to_float(tf.size(mask)) / tf.reduce_sum(mask),
                             false_fn=lambda: output)
            return output

        if training is None:
            training = K.learning_phase()
        output = tf.cond(tf.logical_or(tf.logical_not(training), tf.equal(self.keep_prob, 1.0)),
                         true_fn=lambda: inputs,
                         false_fn=drop)
        return output

    def set_keep_prob(self, keep_prob=None):
        """This method only supports Eager Execution"""
        if keep_prob is not None:
            self.keep_prob = keep_prob
        d, w, h = tf.compat.v1.to_float(self.d), tf.compat.v1.to_float(self.w), tf.compat.v1.to_float(self.h)
        self.gamma = ((1. - self.keep_prob) * (d * w * h) / (self.block_size ** 3) /
                      ((d - self.block_size + 1) * (w - self.block_size + 1) * (h - self.block_size + 1)))

    def _create_mask(self, input_shape):
        sampling_mask_shape = tf.stack([input_shape[0],
                                        self.d - self.block_size + 1,
                                        self.h - self.block_size + 1,
                                        self.w - self.block_size + 1,
                                        self.channel])
        mask = _bernoulli(sampling_mask_shape, self.gamma)
        mask = tf.pad(mask, self.padding)
        mask = tf.nn.max_pool3d(mask, [1, self.block_size, self.block_size, self.block_size, 1], [1, 1, 1, 1, 1], 'SAME')
        mask = 1 - mask
        return mask
```

```
input_layer = layers.Input(shape = (im_height, im_width, im_depth, im_channel))
out1 = layers.Conv3D(filters=8, kernel_size=(3,3,7), activation='relu',input_shape=(im_height, im_width, im_depth, im_channel),padding='s
out1 = DropBlock3D(0.7,3)(out1)
out2 = layers.Conv3D(filters=16, kernel_size=(3,3,5), activation='relu',padding='same')(out1)
out2 = DropBlock3D(0.7,3)(out2)
out3 = layers.Conv3D(filters=32, kernel_size=(3,3,3), activation= 'relu')(out2)
out3 = layers.Reshape((out3.shape[1], out3.shape[2], out3.shape[3]*out3.shape[4]))(out3)
out3 = layers.Conv2D(filters=64, kernel_size=(3,3), activation='relu')(out3)
out4 = layers.Flatten()(out3)
out4 = layers.Dropout(0.4)(out4, training=True)
out4 = layers.Dense(256, activation='relu')(out4)
out5 = layers.Dropout(0.4)(out4,training=True)
out5 = layers.Dense(128, activation='relu')(out5)
#out5 = layers.Dropout(0.4)(out5, training=True)
#out6 = layers.Dense(16, activation='relu')(out5)
model = Model(inputs=input_layer,outputs=out5)
model.summary()
```

```
WARNING:tensorflow:From /usr/local/lib/python3.9/dist-packages/tensorflow/python/util/dispatch.py:1176: to_float (from tensorflow.p
Instructions for updating:
Use `tf.cast` instead.
WARNING:tensorflow:From /usr/local/lib/python3.9/dist-packages/tensorflow/python/autograph/pyct/static_analysis/liveness.py:83: Ana
Instructions for updating:
Lambda fuctions will be no more assumed to be used in the statement where they are used, or at least in the same block. https://git
Model: "model"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_1 (InputLayer)        [(None, 11, 11, 30, 1)]   0

 conv3d (Conv3D)             (None, 11, 11, 30, 8)     512

 drop_block3d (DropBlock3D)  (None, 11, 11, 30, 8)     0

 conv3d_1 (Conv3D)           (None, 11, 11, 30, 16)    5776

 drop_block3d_1 (DropBlock3D  (None, 11, 11, 30, 16)   0
 )

 conv3d_2 (Conv3D)           (None, 9, 9, 28, 32)      13856
```

```
 reshape (Reshape)           (None, 9, 9, 896)         0

 conv2d (Conv2D)             (None, 7, 7, 64)          516160

 flatten (Flatten)           (None, 3136)              0

 dropout (Dropout)           (None, 3136)              0

 dense (Dense)               (None, 256)               803072

 dropout_1 (Dropout)         (None, 256)               0

 dense_1 (Dense)             (None, 128)               32896

=================================================================
Total params: 1,372,272
Trainable params: 1,372,272
Non-trainable params: 0
_________________________________________________________________
```

```
class Prototypical(Model):
    def __init__(self, model, w, h, d, c):
        super(Prototypical, self).__init__()
        self.w, self.h, self.d, self.c = w, h, d, c
        self.encoder = model

    def call(self, support, query, support_labels, query_labels, K, C, N,n_times,training=True):
        n_class = C                                                          #10
        n_support = K                                                        #5
        n_query = N                                                          #15

        if training == True :
            loss = 0
            mc_predictions = []                                              #list of predictions for multiple passes
            for i in range(n_times):
                y = np.zeros((int(C*N),C))                                   #(150,10)
                for i in range(int(C*N)):
                    x = support_labels.index(query_labels[i])                #creation of 1-hot for true labels
                    y[i][x] = 1                                              #n_times passing every query image for calculating variance
                cat = tf.concat([support,query], axis=0)                     #[200,9,9,20,1])
                z = self.encoder(cat)                                        #[200, 320]
                #Divide embedding into support and query
                z_prototypes = tf.reshape(z[:n_class * n_support],[n_class, n_support, z.shape[-1]])   #[10, 5, 320])
                #Prototypes are means of n_support examples
                z_prototypes = tf.math.reduce_mean(z_prototypes, axis=1)     #[10, 320]
                z_query = z[n_class * n_support:]                            #[150, 320]
                #Calculate distances between query and prototypes
                dists = calc_euclidian_dists(z_query, z_prototypes)          #[150, 10]
                #log softmax of calculated distances
                log_p_y = tf.nn.log_softmax(-dists, axis=-1)                 #[150, 10]
                loss1 = -tf.reduce_mean((tf.reduce_sum(tf.multiply(y, log_p_y), axis=-1)))   #loss for the current pass
                loss += loss1                                                #adding loss for each pass
                softmax_predictions = tf.nn.softmax(-dists, axis=-1)         #prediction probability for the search-space classes per
                new_distance = dists/1.8                                     #T = 1.1 to 3.0 for Temperature Scaling
                predictions = tf.nn.softmax(-new_distance, axis=-1)          #prediction probability with temperature scaling
                #print('Loop Number :', i)
                #print("A" * 10)
                #print("Dists : ", dists[0])
                #print("B" * 10)
                #print('New Distance (T applied): ', new_distance[0])
                #print("C" * 10)
                #print("Softmax Predictions : ", softmax_predictions[0])
                #print("D" * 10)
                #print("Temperature Scaling Predictions : ", predictions[0])
                #print()
                mc_predictions.append(predictions)
            y = np.zeros((int(C*N),C))
            for i in range(int(C*N)):
                x = support_labels.index(query_labels[i])                    #creation of 1-hot for true labels
                y[i][x] = 1
            mc_predictions = tf.convert_to_tensor(np.reshape(np.asarray(mc_predictions),(n_times,int(C*N),C)))  #(n_times,150,10)
            std_predictions = tf.math.reduce_std(mc_predictions,axis=0)
            std = tf.reduce_sum(tf.reduce_sum(tf.multiply(std_predictions,y),axis=1))
            #print('std', std)
            loss += mc_loss_weight*std
            temp_list.append(std)
            #calculating mean accuracy
            mean_predictions = tf.reduce_mean(mc_predictions,axis=0)        #mean prediction probability for each class (150,10)
            mean_eq = tf.cast(tf.equal(                                      #accuracy for the current pass
                tf.cast(tf.argmax(mean_predictions, axis=-1), tf.int32),
                tf.cast(tf.argmax(y,axis=-1), tf.int32)), tf.float32)
            mean_accuracy = tf.reduce_mean(mean_eq)
            mean_predictions = tf.reduce_mean(mc_predictions,axis=0)        #mean prediction probability for each class (150,10)
```

```
        mean_predictions = tf.reduce_mean(mc_predictions,axis=0)              #mean prediction probability for each class (150,10)
        return loss, mean_accuracy, mean_predictions
    if training == False :
        loss = 0
        mc_predictions = []                                                   #list of predictions for multiple passes
        for i in range(n_times):                                              #n_times passing the query images for variance calculati
            y = np.zeros((int(C*N),C))                                        #(150,10)
            for i in range(int(C*N)):
                x = support_labels.index(query_labels[i])                     #creation of 1-hot for the true labels
                y[i][x] = 1.
            #merge support and query to forward through encoder
            cat = tf.concat([support,query], axis=0)                          #[200,9,9,20,1]
            z = self.encoder(cat)                                             #[200, 320]
            #Divide embedding into support and query
            z_prototypes = tf.reshape(z[:n_class * n_support],[n_class, n_support, z.shape[-1]])   #[10, 5, 320])
            #Prototypes are means of n_support examples
            z_prototypes = tf.math.reduce_mean(z_prototypes, axis=1)          #[10, 320]
            z_query = z[n_class * n_support:]                                 #[150, 320]
            #Calculate distances between query and prototypes
            dists = calc_euclidian_dists(z_query, z_prototypes)               #[150, 10]
            #log softmax of calculated distances
            log_p_y = tf.nn.log_softmax(-dists, axis=-1)                      #[150, 10]
            loss1 = -tf.reduce_mean((tf.reduce_sum(tf.multiply(y, log_p_y), axis=-1)))
            loss += loss1
            softmax_predictions = tf.nn.softmax(-dists, axis=-1)              #prediction probability for the search-space classes per
            new_distance = dists/1.8                                          #T = 1.1 to 3.0 for Temperature Scaling
            predictions = tf.nn.softmax(-new_distance, axis=-1)               #prediction probability with temperature scaling
            #print('Loop Number :', i)
            #print("A" * 10)
            #print("Dists : ", dists[0])
            #print("B" * 10)
            #print('New Distance (T applied): ', new_distance[0])
            #print("C" * 10)
            #print("Softmax Predictions : ", softmax_predictions[0])
            #print("D" * 10)
            #print("Temperature Scaling Predictions : ", predictions[0])
            mc_predictions.append(predictions)
        y = np.zeros((int(C*N),C))                                            #(150,10)
        for i in range(int(C*N)):
            x = support_labels.index(query_labels[i])                         #creation of 1-hot for the true labels
            y[i][x] = 1.
        mean_predictions = tf.reduce_mean(mc_predictions,axis=0)              #mean prediction probability for each class (150,10)
        mean_eq = tf.cast(tf.equal(                                           #accuracy for the current pass
            tf.cast(tf.argmax(mean_predictions, axis=-1), tf.int32),
            tf.cast(tf.argmax(y,axis=-1), tf.int32)), tf.float32)
        mean_accuracy = tf.reduce_mean(mean_eq)
        mean_pred_index = tf.argmax(mean_predictions,axis=1)
        #mean class-wise accuracies
        mean_correct_class = [[] for i in range(tC)]
        mean_correct_pred = [[] for i in range(tC)]
        classwise_mean_acc = [[] for i in range(tC)]
        for i in range(int(C*N)):
            x = support_labels.index(query_labels[i])
            mean_correct_class[x].append('4')
            if(mean_pred_index[i] == x):
                mean_correct_pred[x].append('4')
        for i in range(tC):
            z = len(mean_correct_pred[i])/len(mean_correct_class[i])
            classwise_mean_acc[i].append(z)
        #std calculation
        std = 0
        for i in range(int(C*N)):
            x = support_labels.index(query_labels[i])
            p_i = np.array([p[i,:] for p in mc_predictions])
            std_i = tf.math.reduce_std(p_i,axis=0)
            std_i_true = std_i[x]
            std += std_i_true                                                 #adding std of each class
        #print('std',std)
        loss += mc_loss_weight*std
        y = np.zeros((int(C*N),C))                                            #(150,10)
        for i in range(int(C*N)):
            x = support_labels.index(query_labels[i])                         #creation of 1-hot for the true labels
            y[i][x] = 1.
        return loss, mc_predictions, mean_accuracy, classwise_mean_acc, y


    def save(self, model_path):
        self.encoder.save(model_path)

    def load(self, model_path):
        self.encoder(tf.zeros([1, self.w, self.h, self.c]))
        self.encoder.load_weights(model_path)
```

Data Loading

```
def loadData(name):
    data = sio.loadmat('/content/drive/My Drive/Data/indian_pines_corrected.mat')['indian_pines_corrected']
    labels = sio.loadmat('/content/drive/My Drive/Data/indian_pines_gt.mat')['indian_pines_gt']
    return data, labels
# without reduction of 200 channels to 30 channels, memory error while creating cube
def applyPCA(X, numComponents):
    newX = np.reshape(X, (-1, X.shape[2]))
    pca = PCA(n_components=numComponents, whiten=True)
    newX = pca.fit_transform(newX)
    newX = np.reshape(newX, (X.shape[0],X.shape[1], numComponents))
    return newX, pca

def padWithZeros(X, margin):
    newX = np.zeros((X.shape[0] + 2 * margin, X.shape[1] + 2* margin, X.shape[2]))
    x_offset = margin
    y_offset = margin
    newX[x_offset:X.shape[0] + x_offset, y_offset:X.shape[1] + y_offset, :] = X
    return newX

def createImageCubes(X, y, windowSize, removeZeroLabels = True):
    margin = int((windowSize - 1) / 2)
    zeroPaddedX = padWithZeros(X, margin=margin)  # X :(145, 145, 30) --> (195, 195, 30) with window =25
    # split patches
    patchesData = np.zeros((X.shape[0] * X.shape[1], windowSize, windowSize, X.shape[2]))  # (21025, 25, 25, 30)
    patchesLabels = np.zeros((X.shape[0] * X.shape[1]))  # (21025,)
    patchIndex = 0

    for r in range(margin, zeroPaddedX.shape[0] - margin):
        for c in range(margin, zeroPaddedX.shape[1] - margin):
            patch = zeroPaddedX[r - margin:r + margin + 1, c - margin:c + margin + 1]
            patchesData[patchIndex, :, :, :] = patch
            patchesLabels[patchIndex] = y[r-margin, c-margin]
            patchIndex = patchIndex + 1

    patchesData = np.expand_dims(patchesData, axis=-1)
    return patchesData,patchesLabels


dataset1 = 'indian_pines'                                          # 16 classes
ip_x1, ip_y = loadData(dataset1)                                 #((512, 217, 204), (512, 217))
ip_x2,pca = applyPCA(ip_x1,numComponents=30)                      # ((512, 217, 20), (512, 217))
ip_X,ip_Y = createImageCubes(ip_x2, ip_y, windowSize=11)   #(111104, 9, 9, 20, 1) (111104,)
print(ip_X.shape,ip_Y.shape)
```

```
(21025, 11, 11, 30, 1) (21025,)
```

```
def patches_class(X,Y,n) :
  n_classes = n
  patches_list = []
  for i in range(1,n_classes+1):   # not considering class 0
    patchesData_Ith_Label = X[Y==i,:,:,:,:]
    patches_list.append(patchesData_Ith_Label)
  return patches_list


patches_class_ip = patches_class(ip_X,ip_Y,16) # class_wise list of patches #(16,) for class 0: (2009, 9, 9, 20, 1)


train_class_indices = [1,2,4,5,7,9,10,11,13,14]
test_class_indices = [0,3,6,8,12,15]
train_patches_class = [patches_class_ip[i] for i in train_class_indices]        #(10)
test_patches_class = [patches_class_ip[i] for i in test_class_indices]        #(6)
train_class_labels = [2,3,5,6,8,10,11,12,14,15]
test_class_labels = [1,4,7,9,13,16]     #[11...16]
```

Prepare Training and validation Dataset

Create DataLoader

```
C = 5  # n_class
K1 = 5   # n_support
N = 15   # n_query
tC = 3   # classes in a test episode
im_height,im_width,im_depth = 11,11,30
```

```
def new_episode(patches_list,K,C,N,class_labels) :
  selected_classes = np.random.choice(class_labels,C,replace=False)  # Randomly choice 5 Classes out of classes available
  tsupport_patches = []
  tquery_patches = []
  query_labels = []
  support_labels = list(selected_classes)
  for x in selected_classes :
    sran_indices = np.random.choice(len(patches_list[x-1]),K,replace=False)  # for class no X-1: select random sample no
    support_patches = patches_list[x-1][sran_indices,:,:,:,:]
    qran_indices = np.random.choice(len(patches_list[x-1]),N,replace=False)  # N Samples for Query
    query_patches = patches_list[x-1][qran_indices,:,:,:,:]
  # Support and Query patches belong to same Class
    for i in range(N) :
      query_labels.append(x)    # N Samples for Query
    tquery_patches.extend(query_patches)
    tsupport_patches.extend(support_patches)
  temp1 = list(zip(tquery_patches, query_labels))
  random.shuffle(temp1)        # By Doing Shuffling, Support, Query Same class combination got mismatched - mitigated by support index
  tquery_patches, query_labels = zip(*temp1)
  tquery_patches = tf.convert_to_tensor(np.reshape(np.asarray(tquery_patches),(C*N,im_height,im_width,im_depth,1)),dtype=tf.float32)
  tsupport_patches = tf.convert_to_tensor(np.reshape(np.asarray(tsupport_patches),(C*K,im_height,im_width,im_depth,1)),dtype=tf.float32)
  return tquery_patches, tsupport_patches, query_labels, support_labels


tquery_patches, tsupport_patches, query_labels, support_labels = new_episode(patches_class_ip,K1,C,N,train_class_labels)
```

Training

```
ProtoModel = Prototypical(model,im_width, im_height, im_depth, im_channel)
optimizer = tf.keras.optimizers.Adam(0.00001)          #Adam(0.001)
n_times = 10
#std_list = []




# Metrics to gather
train_loss = tf.metrics.Mean(name='train_loss')
train_acc = tf.metrics.Mean(name='train_accuracy')

def train_step(support, query, support_labels, query_labels, K, C, N):
    # Forward & update gradients
    with tf.GradientTape() as tape:
        loss, mean_accuracy, mean_predictions = ProtoModel(support, query, support_labels, query_labels, K, C, N,n_times,training=True)
    gradients = tape.gradient(loss, model.trainable_variables)
    optimizer.apply_gradients(zip(gradients, model.trainable_variables))

    # Log loss and accuracy for step
    train_loss(loss)
    train_acc(mean_accuracy)


checkpoint_dir = '/content/drive/My Drive/Research/SAMPLE_CODE_TF2_Keras/Inprogress/Hyperspectral_Classification/Training checkpoints/IP/
checkpoint_prefix = os.path.join(checkpoint_dir, "ckpt")
checkpoint = tf.train.Checkpoint(optimizer=optimizer,
                                 ProtoModel = ProtoModel)


n_episodes = 100
temp_list = []
for epoch in range(50): # n_epochs-140
    train_loss.reset_states()
    train_acc.reset_states()

    for epi in range(n_episodes):
        tquery_patches, tsupport_patches, query_labels, support_labels = new_episode(patches_class_ip,K1,C,N,train_class_labels)
        train_step(tsupport_patches, tquery_patches,support_labels, query_labels, K1, C, N)
        template = 'Epoch {}, Episode {}, Train Loss: {:.2f}, Train Accuracy: {:.2f}'
        print(template.format(epoch+1, epi+1,train_loss.result(),train_acc.result()*100))
    if epoch % 5 == 0 and epoch != 0 :
        checkpoint.save(file_prefix = checkpoint_prefix)
```

```
Epoch 50, Episode 53, Train Loss: 25.95, Train Accuracy: 97.48
Epoch 50, Episode 54, Train Loss: 26.05, Train Accuracy: 97.46
Epoch 50, Episode 55, Train Loss: 25.90, Train Accuracy: 97.50
Epoch 50, Episode 56, Train Loss: 26.03, Train Accuracy: 97.55
Epoch 50, Episode 57, Train Loss: 25.88, Train Accuracy: 97.57
Epoch 50, Episode 58, Train Loss: 25.59, Train Accuracy: 97.61
Epoch 50, Episode 59, Train Loss: 25.56, Train Accuracy: 97.65
Epoch 50, Episode 60, Train Loss: 25.52, Train Accuracy: 97.64
Epoch 50, Episode 61, Train Loss: 25.64, Train Accuracy: 97.55
Epoch 50, Episode 62, Train Loss: 25.87, Train Accuracy: 97.57
Epoch 50, Episode 63, Train Loss: 25.65, Train Accuracy: 97.61
Epoch 50, Episode 64, Train Loss: 25.49, Train Accuracy: 97.65
Epoch 50, Episode 65, Train Loss: 25.59, Train Accuracy: 97.58
Epoch 50, Episode 66, Train Loss: 25.44, Train Accuracy: 97.62
Epoch 50, Episode 67, Train Loss: 25.41, Train Accuracy: 97.65
Epoch 50, Episode 68, Train Loss: 25.35, Train Accuracy: 97.65
Epoch 50, Episode 69, Train Loss: 25.20, Train Accuracy: 97.68
Epoch 50, Episode 70, Train Loss: 25.30, Train Accuracy: 97.71
Epoch 50, Episode 71, Train Loss: 25.37, Train Accuracy: 97.75
Epoch 50, Episode 72, Train Loss: 25.25, Train Accuracy: 97.78
Epoch 50, Episode 73, Train Loss: 25.30, Train Accuracy: 97.81
Epoch 50, Episode 74, Train Loss: 25.41, Train Accuracy: 97.77
Epoch 50, Episode 75, Train Loss: 25.37, Train Accuracy: 97.78
Epoch 50, Episode 76, Train Loss: 25.49, Train Accuracy: 97.70
Epoch 50, Episode 77, Train Loss: 25.48, Train Accuracy: 97.71
Epoch 50, Episode 78, Train Loss: 25.50, Train Accuracy: 97.73
Epoch 50, Episode 79, Train Loss: 25.70, Train Accuracy: 97.72
Epoch 50, Episode 80, Train Loss: 25.72, Train Accuracy: 97.72
Epoch 50, Episode 81, Train Loss: 25.73, Train Accuracy: 97.74
Epoch 50, Episode 82, Train Loss: 25.58, Train Accuracy: 97.77
Epoch 50, Episode 83, Train Loss: 25.45, Train Accuracy: 97.80
Epoch 50, Episode 84, Train Loss: 25.48, Train Accuracy: 97.79
Epoch 50, Episode 85, Train Loss: 25.36, Train Accuracy: 97.82
Epoch 50, Episode 86, Train Loss: 25.36, Train Accuracy: 97.81
Epoch 50, Episode 87, Train Loss: 25.53, Train Accuracy: 97.79
Epoch 50, Episode 88, Train Loss: 25.55, Train Accuracy: 97.76
Epoch 50, Episode 89, Train Loss: 25.40, Train Accuracy: 97.78
Epoch 50, Episode 90, Train Loss: 25.38, Train Accuracy: 97.81
Epoch 50, Episode 91, Train Loss: 25.45, Train Accuracy: 97.79
Epoch 50, Episode 92, Train Loss: 25.41, Train Accuracy: 97.81
Epoch 50, Episode 93, Train Loss: 25.42, Train Accuracy: 97.82
Epoch 50, Episode 94, Train Loss: 25.34, Train Accuracy: 97.84
Epoch 50, Episode 95, Train Loss: 25.42, Train Accuracy: 97.81
Epoch 50, Episode 96, Train Loss: 25.44, Train Accuracy: 97.81
Epoch 50, Episode 97, Train Loss: 25.44, Train Accuracy: 97.81
Epoch 50, Episode 98, Train Loss: 25.48, Train Accuracy: 97.81
Epoch 50, Episode 99, Train Loss: 25.58, Train Accuracy: 97.83
```

Tuning_5

```
tune_set_5 = [[] for i in range(6)]
for j in range(6) :
  tune_set_5[j] = test_patches_class[j][:5,:,:,:,:]   # for each class first 5 samples taken
std_tune_5 = []
```

```
def tune_episode(tune_set,tC,tK,tN,test_class_labels) :
  selected_classes = np.random.choice(test_class_labels,tC,replace=False)
  support_labels  = list(selected_classes)
  query_labels = []
  support_patches = []
  query_patches = []
  for x in selected_classes :
    y = test_class_labels.index(x)
    np.random.shuffle(tune_set[y])
    support_imgs = tune_set[y][:tK,:,:,:,:]    #Support 1, Query 4
    query_imgs = tune_set[y][tK:5,:,:,:,:]
    support_patches.extend(support_imgs)
    query_patches.extend(query_imgs)
    for i in range(tN) :
      query_labels.append(x)
  temp1 = list(zip(query_patches, query_labels))
  random.shuffle(temp1)
  query_patches, query_labels = zip(*temp1)
  query_patches = tf.convert_to_tensor(np.reshape(np.asarray(query_patches),(tC*tN,im_height,im_width,im_depth,1)),dtype=tf.float32)
  support_patches = tf.convert_to_tensor(np.reshape(np.asarray(support_patches),(tC*tK,im_height,im_width,im_depth,1)),dtype=tf.float32)
  return query_patches, support_patches, query_labels, support_labels


query_patches, support_patches, query_labels, support_labels = tune_episode(tune_set_5,4,1,4,test_class_labels)


# Metrics to gather
tune_loss = tf.metrics.Mean(name='tune_loss')
tune_acc = tf.metrics.Mean(name='tune_accuracy')
```

```
def tune_step(support, query, support_labels, query_labels, K, C, N):
    # Forward & update gradients
    with tf.GradientTape() as tape:
        loss, mean_accuracy, mean_predictions = ProtoModel(support, query, support_labels, query_labels, K, C, N,n_times,training=True)
    gradients = tape.gradient(loss, model.trainable_variables)
    optimizer.apply_gradients(zip(gradients, model.trainable_variables))

    # Log loss and accuracy for step
    tune_loss(loss)
    tune_acc(mean_accuracy)

checkpoint_dir1 = '/content/drive/My Drive/Research/SAMPLE_CODE_TF2_Keras/Inprogress/Hyperspectral_Classification/Training checkpoints/IP
checkpoint_prefix1 = os.path.join(checkpoint_dir1, "ckpt")
checkpoint = tf.train.Checkpoint(optimizer=optimizer,
                                 ProtoModel = ProtoModel)


n_episodes = 100
temp_list = []
std_tune_5 = []
for epoch in range(41):
    tune_loss.reset_states()
    tune_acc.reset_states()
    for epi in range(n_episodes+1):
        tquery_patches, tsupport_patches, query_labels, support_labels = tune_episode(tune_set_5,3,1,4,test_class_labels)
        tune_step(tsupport_patches, tquery_patches,support_labels, query_labels, 1, 3, 4)

    template = 'Epoch {}, Tune Loss: {:.2f}, Tune Accuracy: {:.2f}'
    print(template.format(epoch+1,tune_loss.result(),tune_acc.result()*100))
    if (epoch+1)%5 == 0 :
      checkpoint.save(file_prefix = checkpoint_prefix1)
```

```
Epoch 1, Tune Loss: 5.60, Tune Accuracy: 99.75
Epoch 2, Tune Loss: 2.65, Tune Accuracy: 100.00
Epoch 3, Tune Loss: 1.59, Tune Accuracy: 100.00
Epoch 4, Tune Loss: 1.51, Tune Accuracy: 100.00
Epoch 5, Tune Loss: 1.15, Tune Accuracy: 100.00
Epoch 6, Tune Loss: 1.01, Tune Accuracy: 100.00
Epoch 7, Tune Loss: 0.81, Tune Accuracy: 100.00
Epoch 8, Tune Loss: 0.93, Tune Accuracy: 100.00
Epoch 9, Tune Loss: 0.65, Tune Accuracy: 100.00
Epoch 10, Tune Loss: 0.56, Tune Accuracy: 100.00
Epoch 11, Tune Loss: 0.51, Tune Accuracy: 100.00
Epoch 12, Tune Loss: 0.50, Tune Accuracy: 100.00
Epoch 13, Tune Loss: 0.48, Tune Accuracy: 100.00
Epoch 14, Tune Loss: 0.48, Tune Accuracy: 100.00
Epoch 15, Tune Loss: 0.52, Tune Accuracy: 100.00
Epoch 16, Tune Loss: 0.41, Tune Accuracy: 100.00
Epoch 17, Tune Loss: 0.37, Tune Accuracy: 100.00
Epoch 18, Tune Loss: 0.42, Tune Accuracy: 100.00
Epoch 19, Tune Loss: 0.34, Tune Accuracy: 100.00
Epoch 20, Tune Loss: 0.29, Tune Accuracy: 100.00
Epoch 21, Tune Loss: 0.37, Tune Accuracy: 100.00
Epoch 22, Tune Loss: 0.20, Tune Accuracy: 100.00
Epoch 23, Tune Loss: 0.24, Tune Accuracy: 100.00
Epoch 24, Tune Loss: 0.27, Tune Accuracy: 100.00
Epoch 25, Tune Loss: 0.19, Tune Accuracy: 100.00
Epoch 26, Tune Loss: 0.25, Tune Accuracy: 100.00
Epoch 27, Tune Loss: 0.25, Tune Accuracy: 100.00
Epoch 28, Tune Loss: 0.21, Tune Accuracy: 100.00
Epoch 29, Tune Loss: 0.24, Tune Accuracy: 100.00
Epoch 30, Tune Loss: 0.16, Tune Accuracy: 100.00
Epoch 31, Tune Loss: 0.14, Tune Accuracy: 100.00
Epoch 32, Tune Loss: 0.17, Tune Accuracy: 100.00
Epoch 33, Tune Loss: 0.15, Tune Accuracy: 100.00
Epoch 34, Tune Loss: 0.21, Tune Accuracy: 100.00
Epoch 35, Tune Loss: 0.09, Tune Accuracy: 100.00
Epoch 36, Tune Loss: 0.08, Tune Accuracy: 100.00
Epoch 37, Tune Loss: 0.11, Tune Accuracy: 100.00
Epoch 38, Tune Loss: 0.14, Tune Accuracy: 100.00
Epoch 39, Tune Loss: 0.10, Tune Accuracy: 100.00
Epoch 40, Tune Loss: 0.13, Tune Accuracy: 100.00
Epoch 41, Tune Loss: 0.11, Tune Accuracy: 100.00
```

```
def test_episode(test_patches_class,test_class_labels,test_C,test_K,i,f) :
  selected_classes = test_class_labels[i:f]   # [1, 2, 3, 4, 5, 6, 7, 8]
  support_labels = list(selected_classes)
  query_labels = []
  support_patches = []
  query_patches = []
  for x in selected_classes :
    y = test_class_labels.index(x)
    support_imgs = test_patches_class[y][:test_K,:,:,:,:]
    query_imgs = test_patches_class[y][test_K:,:,:,:,:]
    support_patches.extend(support_imgs)
```

```
    query_patches.extend(query_imgs)
    for i in range(query_imgs.shape[0]) :
      query_labels.append(x)
  temp1 = list(zip(query_patches, query_labels))
  random.shuffle(temp1)
  query_patches, query_labels = zip(*temp1)
  x = len(query_labels)
  query_patches = tf.convert_to_tensor(np.reshape(np.asarray(query_patches),(x,im_height,im_width,im_depth,1)),dtype=tf.float32)
  support_patches = tf.convert_to_tensor(np.reshape(np.asarray(support_patches),(test_C*test_K,im_height,im_width,im_depth,1)),dtype=tf.f
  return query_patches, support_patches, query_labels, support_labels,x
```

```
query_patches, support_patches, query_labels, support_labels,x = test_episode(test_patches_class,test_class_labels,3,5,0,3)
```

```
# Metrics to gather
test_loss = tf.metrics.Mean(name='test_loss')
test_acc = tf.metrics.Mean(name='test_accuracy')
def test_step(support, query, support_labels, query_labels, K, C, y):
    loss, mc_predictions, mean_accuracy, classwise_mean_acc, y = ProtoModel(support, query, support_labels, query_labels, K, C, y,n_times
    return loss, mc_predictions, mean_accuracy, classwise_mean_acc, y
```

```
for epoch in range(1): #1000
    test_loss.reset_states()
    test_acc.reset_states()
    tquery_patches1, tsupport_patches1, query_labels1, support_labels1, x1 = test_episode(test_patches_class,test_class_labels,3,5,0,3)
    loss1, mc_predictions1, mean_accuracy1, classwise_mean_acc1, y1 = test_step(tsupport_patches1, tquery_patches1,support_labels1, query
    print('OA1',mean_accuracy1)
# Class-wise Accuracy
    for i in range(tC) :
      print('class',i+1,classwise_mean_acc1[i])
    print('loss',loss1)
    tquery_patches2, tsupport_patches2, query_labels2, support_labels2, x2 = test_episode(test_patches_class,test_class_labels,3,5,3,6)
    loss2, mc_predictions2, mean_accuracy2, classwise_mean_acc2, y2 = test_step(tsupport_patches2, tquery_patches2,support_labels2, query
    print('OA2',mean_accuracy2)
# Class-wise Accuracy
    for i in range(tC) :
      print('class',i+4,classwise_mean_acc2[i])
    print('loss',loss2)
```

```
OA1 tf.Tensor(0.902027, shape=(), dtype=float32)
class 1 [0.926829268292683]
class 2 [0.8879310344827587]
class 3 [1.0]
loss tf.Tensor(95.889, shape=(), dtype=float32)
OA2 tf.Tensor(0.980198, shape=(), dtype=float32)
class 4 [1.0]
class 5 [0.97]
class 6 [1.0]
loss tf.Tensor(51.91124, shape=(), dtype=float32)
```

Overall Accuracy

```
mean_predictions1 =  tf.reduce_mean(mc_predictions1,axis=0)
mean_predictions2 =  tf.reduce_mean(mc_predictions2,axis=0)
overall_predictions = tf.concat([mean_predictions1,mean_predictions2],axis=0)
overall_true_labels = tf.concat([y1,y2],axis=0)
correct_pred = tf.cast(tf.equal(                                          # accuracy for the current pass
            tf.cast(tf.argmax(overall_predictions, axis=-1), tf.int32),
            tf.cast(tf.argmax(overall_true_labels,axis=-1), tf.int32)), tf.float32)
o_acc = tf.reduce_mean(correct_pred)
print("Overall accuracy:",o_acc)
```

```
Overall accuracy: tf.Tensor(0.94156927, shape=(), dtype=float32)
```

Confusion Matrix

```
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
mean_predictions1 =  tf.reduce_mean(mc_predictions1,axis=0)
cm_pred1 = tf.argmax(mean_predictions1, axis=-1)
mean_predictions2 =  tf.reduce_mean(mc_predictions2,axis=0)
cm_pred2 = tf.argmax(mean_predictions2, axis=-1) + 3
overall_predictions = tf.concat([cm_pred1,cm_pred2],axis=0)
cm_true1 = tf.argmax(y1,axis=-1)
cm_true2 = tf.argmax(y2,axis=-1) + 3
overall_true_labels = tf.concat([cm_true1,cm_true2],axis=0)
results = confusion_matrix(overall_true_labels,overall_predictions)
```

```
print ('Confusion Matrix :')
print(results)
print ('Report : ')
print (classification_report(overall_true_labels, overall_predictions))
```

```
Confusion Matrix :
[[ 38   0   3   0   0   0]
 [ 23 206   3   0   0   0]
 [  0   0  23   0   0   0]
 [  0   0   0  15   0   0]
 [  0   0   0   6 194   0]
 [  0   0   0   0   0  88]]
Report :
              precision    recall  f1-score   support

           0       0.62      0.93      0.75        41
           1       1.00      0.89      0.94       232
           2       0.79      1.00      0.88        23
           3       0.71      1.00      0.83        15
           4       1.00      0.97      0.98       200
           5       1.00      1.00      1.00        88

    accuracy                           0.94       599
   macro avg       0.86      0.96      0.90       599
weighted avg       0.96      0.94      0.95       599
```

Kappa Accuracy

```
import sklearn
sklearn.metrics.cohen_kappa_score(overall_true_labels, overall_predictions)
```

```
0.9196687894184273
```

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